

POLYP DETECTION MEDICAL REPORT

Analysis Date: June 04, 2026

VIDEO INFORMATION

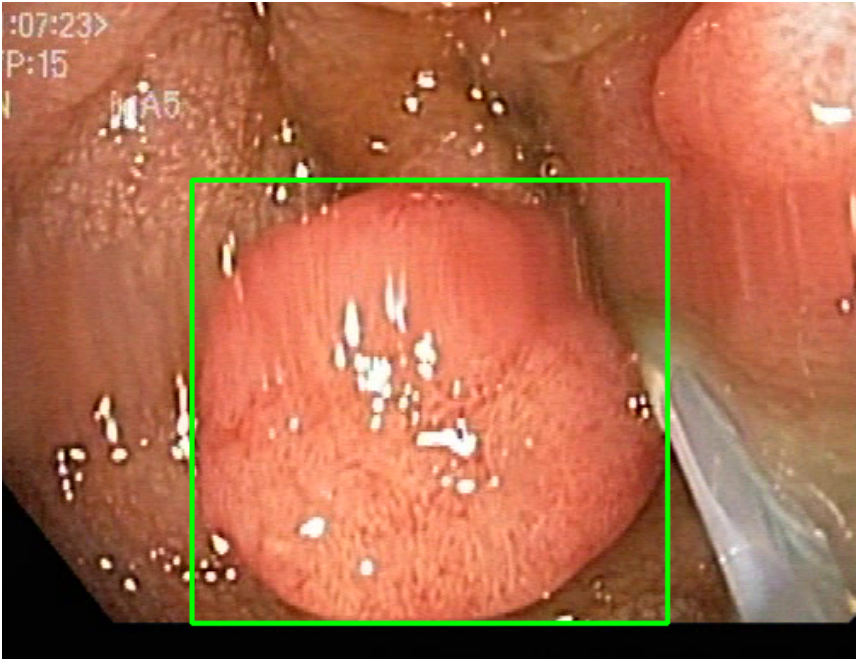
Video ID:	be97d107-e0fd-4d24-bc77-0ce9f88747a4
Total Frames:	3007
Frame Rate:	25.0 fps (assumed)
Duration:	120.3 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	2680	✓
RT-DETR Detections	3048	✓
MedSAM2 Detections	3376	✓
Consensus (3-Model Agreement)	3	✓
Risk Classification	Count	Percentage
HIGH RISK	3	100.0%
MEDIUM RISK	0	0.0%
LOW RISK	0	0.0%
TOTAL ANALYZED	3	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — BLEEDING_POLYP [HIGH_RISK] (Tracked across 258 frames)



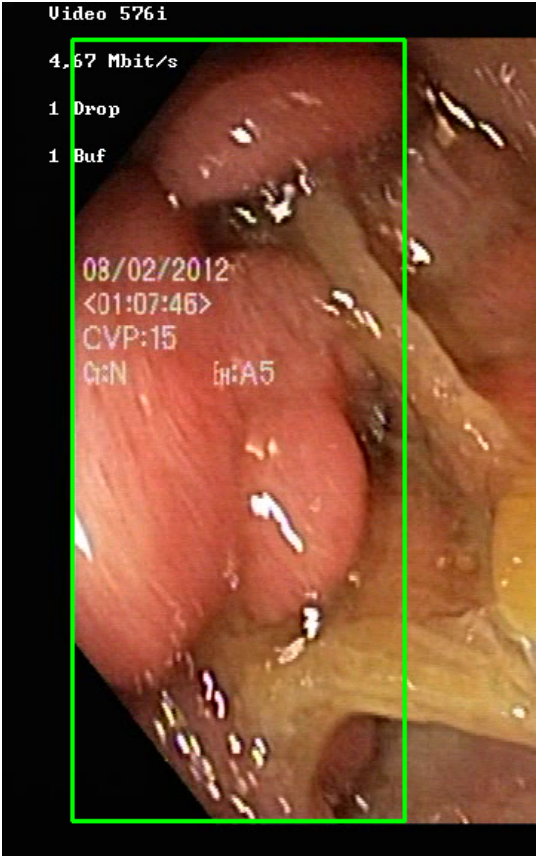
Feature	Value	Interpretation
Frame Index	1820	Frame 1820 of 3007
Detection Method	Detector Consensus (MedSAM2 Unavailable)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	BLEEDING_POLYP	Type confidence: 92.0%
Redness Score	0.208	Color-based vascularization
Relative Radius	35.1%	Polyp size as % of frame diagonal
Texture Score	0.289	Surface morphology
Vessel Visibility	0.898	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output

Clinical Class	MALIGNANT_P OLYP	High suspicion for malignancy — biopsy and urgent resection recommended
Detection Confidence	87.8%	YOLO / RT-DETR / track confidence

Type Assessment: Active hemorrhagic lesion — elevated vascularity with redness signal. Immediate clinical review required.

Low-confidence HIGH RISK detection - Further evaluation recommended

POLYP #2 — MALIGNANT_POLYP [HIGH_RISK] (Tracked across 1592 frames)



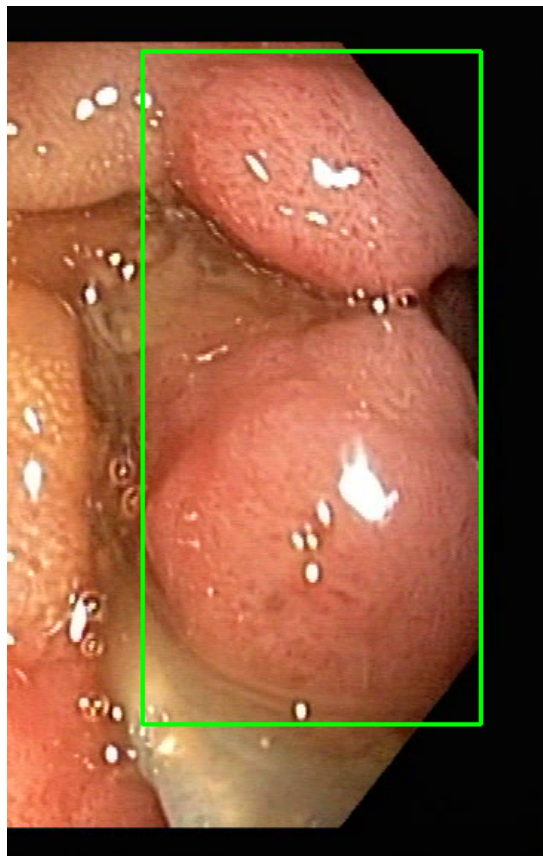
Feature	Value	Interpretation
Frame Index	2412	Frame 2412 of 3007
Detection Method	Detector Consensus (YOLO / RT-DETR / MedSAM2 Unavailable)	Detector Consensus (YOLO / RT-DETR / MedSAM2 Unavailable) (MedSAM2 unavailable for this video)
POLYP TYPE	MALIGNANT_P OLYP	Type confidence: 92.0%
Redness Score	0.252	Color-based vascularization

Relative Radius	32.3%	Polyp size as % of frame diagonal
Texture Score	0.098	Surface morphology
Vessel Visibility	0.410	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	88.4%	YOLO / RT-DETR / track confidence

Type Assessment: Morphology consistent with high-grade neoplasia — biopsy and urgent resection strongly recommended.

Low-confidence HIGH RISK detection - Further evaluation recommended

POLYP #3 — BLEEDING_POLYP [HIGH_RISK] (Tracked across 47 frames)



Feature	Value	Interpretation
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Frame Index	1859	Frame 1859 of 3007
Detection Method	Detector Consensus (MedSAM2 Unavailable)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	BLEEDING_POLYP	Type confidence: 92.0%
Redness Score	0.192	Color-based vascularization
Relative Radius	26.4%	Polyp size as % of frame diagonal
Texture Score	0.226	Surface morphology
Vessel Visibility	0.909	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	LOW_RISK	Final symbolic reasoning bucket
Expert Prediction	LOW_RISK	Decision Tree output
Clinical Class	MALIGNANT_POLYP	High suspicion for malignancy — biopsy and urgent resection recommended
Detection Confidence	85.3%	YOLO / RT-DETR / track confidence

Type Assessment: Active hemorrhagic lesion — elevated vascularity with redness signal. Immediate clinical review required.

Low-confidence HIGH RISK detection - Further evaluation recommended

CLINICAL IMPRESSION

Summary:

A total of 3 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:

- HIGH RISK polyps: 3 (100.0%)
- MEDIUM RISK polyps: 0 (0.0%)
- LOW RISK polyps: 0 (0.0%)

Recommendation:

The presence of HIGH RISK polyps warrants close clinical attention. Consider endoscopic resection or follow-up surveillance depending on clinical context.

Technical Details:

- Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
- Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
- Classification: Cluster-specific Decision Tree experts
- Confidence: Based on deep learning + medical heuristics

Disclaimer: This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.